

GJ 8999 b Exoplanet Characterization

Introduction

In this assignment, we will estimate the mass and radius of the exoplanet GJ 8999 b, which is orbiting the M dwarf star GJ 8999.

Star Properties

The known properties of the star GJ 8999 are listed below:

- Mass of GJ 8999 (M_{star}): $0.2 M_{\odot}$
- Radius of GJ 8999 (R_{star}): $0.2 R_{\odot}$

1. Characterizing GJ 8999 b using Transit Data

The transit method is a key technique for detecting and characterizing exoplanets. It relies on observing the slight decrease in a star's brightness as a planet passes directly in front of it from our vantage point.

1.1 Transit Depth

The fractional decrease in the star's brightness during a transit is known as the transit depth. This depth is directly related to the ratio of the planet's radius to the star's radius. The transit depth can be expressed as:

$$\text{Transit Depth} = \left(\frac{R_{\text{planet}}}{R_{\text{star}}} \right)^2$$

From the measured transit depth, the radius of the planet (R_{planet}) can be estimated by rearranging this formula:

$$R_{\text{planet}} = R_{\text{star}} \sqrt{\text{Transit Depth}}$$

2. Characterizing GJ 8999 b using Radial Velocity Data

The radial velocity method detects exoplanets by observing the gravitational tug a planet exerts on its host star. This tug causes the star to wobble slightly, resulting in a detectable shift in its spectral lines due to the Doppler effect. The amplitude of this wobble, known as the radial velocity amplitude (K), is directly related to the planet's mass.

The radial velocity amplitude (K) is given by the following equation, which accounts for the planet's mass (M_p), orbital period (P), the star's mass ($M_{\star}$), and the inclination (i) of the orbital plane:

$$K = M_p \sin i \left(\frac{2\pi G}{PM_\star^2} \right)^{1/3}$$

To determine the planet's mass (M_p), this equation can be rearranged as follows:

$$M_p = K \frac{1}{\sin i} \left(\frac{PM_\star^2}{2\pi G} \right)^{1/3}$$

It's important to note that the radial velocity method inherently provides the *minimum* mass of the planet, $M_p \sin i$, because the inclination i is usually unknown from radial velocity data alone. If the inclination can be determined from the transit method (where a transit implies an inclination close to 90°), then the true mass (M_p) can be calculated.

Conclusion

By utilizing both the transit method to determine the planet's radius and potentially its inclination, and the radial velocity method to derive its mass (or minimum mass), we can comprehensively characterize the mysterious exoplanet GJ 8999 b. The combination of these techniques provides the necessary data to publish a complete study of this exoplanetary system.
